

A Duty-Cycled PLL Architecture for a Low-Power IR-UWB Transmitter in Neural Implants

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Abstract—We present in this paper a low-power phase-locked loop (PLL) architecture with tunable duty-cycle operation for data transmission from implantable neural recording devices. By integrating a switch-controlled voltage holding (SCVH) module, the architecture enables periodic calibration, while ensuring the voltage drop during silent phases is negligible. Implemented in TSMC 40 nm CMOS process technology, the core area is 0.0035 mm², and the post-layout simulation reveals that the proposed PLL achieves a locking frequency of 4.25 GHz in 0.5 μ s and a faster relocking in subsequent calibration cycles. The average power consumption is 90 μ W, reduced by 89% compared to the design without the proposed SCVH module, and the peak-to-peak jitter is 2.14 ps.

Index Terms—Neural implants, Brain-machine interfaces, Wireless transmitter, IR-UWB, Low power, PLL, Duty-cycle mode.

I. INTRODUCTION

Neural implants are electronic devices surgically implanted into the nervous system to restore, enhance, or replace impaired neural functions. They have been successfully applied in various medical domains and hold great promise for future treatments of complex neurological disorders such as Alzheimer's disease, depression, and for the restoration of motor function in paralyzed patients. As illustrated in Fig. 1, a typical neural implant system comprises an analog front-end (AFE) for neural signal acquisition and a data transmitter to wirelessly send information to external devices. To minimize invasiveness and infection risk while enabling patient mobility, non-invasive wireless transmitters have become a more favorable solution [1]–[7].

The transmitter receives digitized neural signals, upconverts them, and amplifies the power before feeding them to an antenna. A stable carrier frequency is typically generated using either an oscillator or a phase-locked loop (PLL). Conventional open-loop oscillators, which rely heavily on external control voltages, are highly susceptible to power supply noise, temperature variations, and process deviations [8], [9]. These factors can cause long-term carrier frequency drift, eventually violating spectrum regulations. In contrast, PLLs demonstrate superior performance in suppressing frequency drift and stabilizing the carrier signal. Analog PLLs are characterized by superior phase noise performance and the absence of quantization noise compared to digital PLLs. And for low-power wireless transmission in implants, the analog implementation avoids the high-frequency sampling

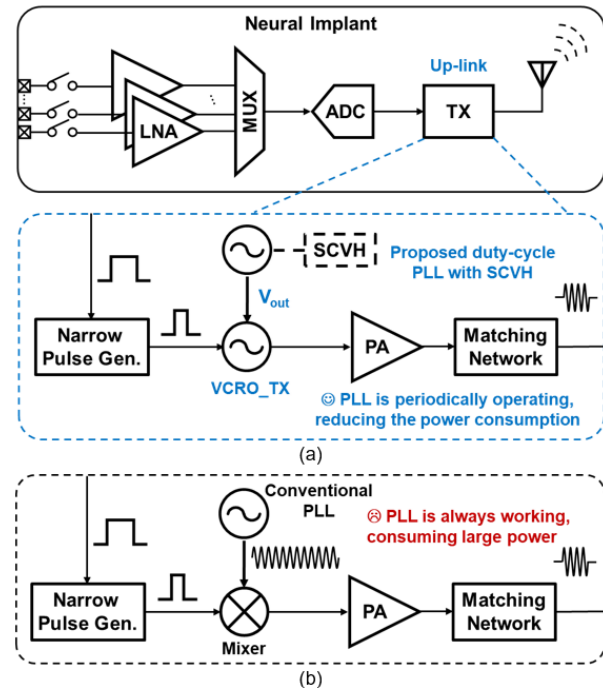


Fig. 1. (a) Block diagram of a neural implant including an impulse-radio ultra-wideband (IR-UWB) wireless transmitter, and proposed hybrid PLL+VCRO_TX scheme. (b) Conventional PLL scheme for upconversion.

clock required by complex digital loop filters and ensures a cleaner transmitter output spectrum [10], [11]. However, due to output oscillation for the mixer, the conventional PLL used for upconversion often needs to remain continuously active, and consume mW-level power [12]–[15]. This leads to significantly higher power consumption, compared to switchable oscillators that can enter a sleep mode when not transmitting data, thereby limiting the operational lifespan of the neural implant [16], [17].

To address the dual challenges of low power consumption and a stable carrier frequency in neural implants, this paper proposes a hybrid PLL-VCRO upconversion scheme. The scheme utilizes a primary phase-locked loop (PLL) to generate a stable control voltage, which is then applied to a second, identical voltage-controlled ring oscillator (VCRO_TX) dedicated to the transmitter's up-conversion process. By leveraging the slow and continuous nature of environmental changes inside the human body, the proposed method enables periodic

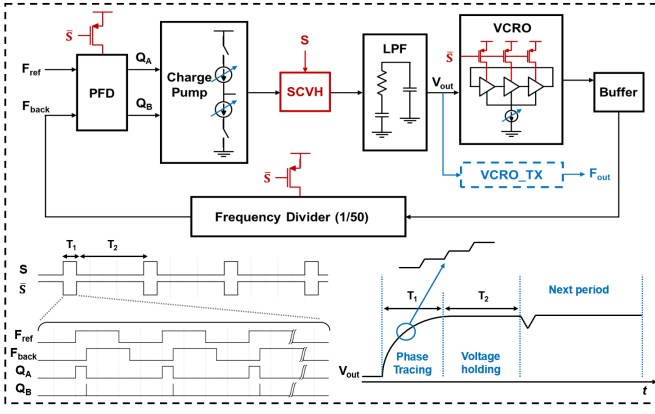


Fig. 2. Architecture and timing diagram of proposed PLL.

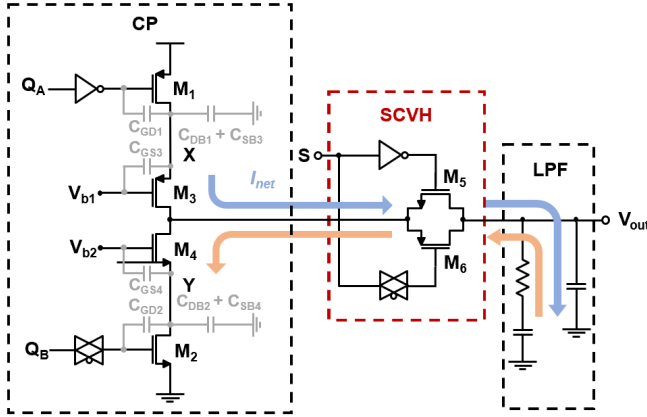


Fig. 3. Circuit implementation of proposed CP, SCVH, and LPF.

carrier frequency calibration. A duty-cycled calibration-silence operation mode is designed to significantly reduce power consumption while maintaining a stable carrier frequency, thus extending the implant's operational life without compromising transmission performance.

The remaining parts of this paper provide a comprehensive overview of the proposed PLL architecture and circuit implementation in Section II. The post-layout simulation results are presented in Section III, followed by a summary of the contributions of this work in Section IV.

II. PROPOSED LOW-POWER PLL DESIGN

A. Low-Power PLL Architecture

The proposed low-power PLL architecture, illustrated in Fig. 2, employs a novel dual-oscillator scheme to achieve duty-cycled operation. The system comprises a phase-frequency detector (PFD), a charge pump (CP), a low-pass filter (LPF), a frequency divider, a switch-controlled voltage holding (SCVH) module, and two identical voltage-controlled ring oscillators (VCROs). One VCRO operates within the PLL's feedback loop, while the second (VCRO_TX) is dedicated to the transmitter's up-conversion stage.

The PFD detects whether the reference signal F_{ref} leads or lags the feedback signal F_{back} , generating corresponding

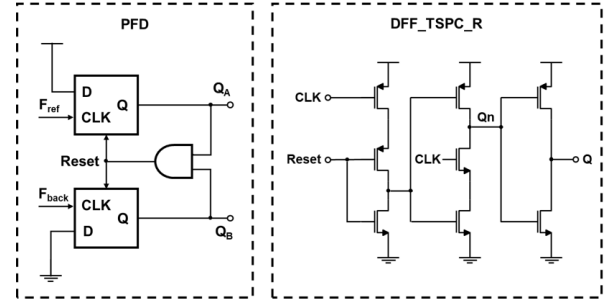


Fig. 4. Circuit implementation of PFD and DFF_TSPC_R.

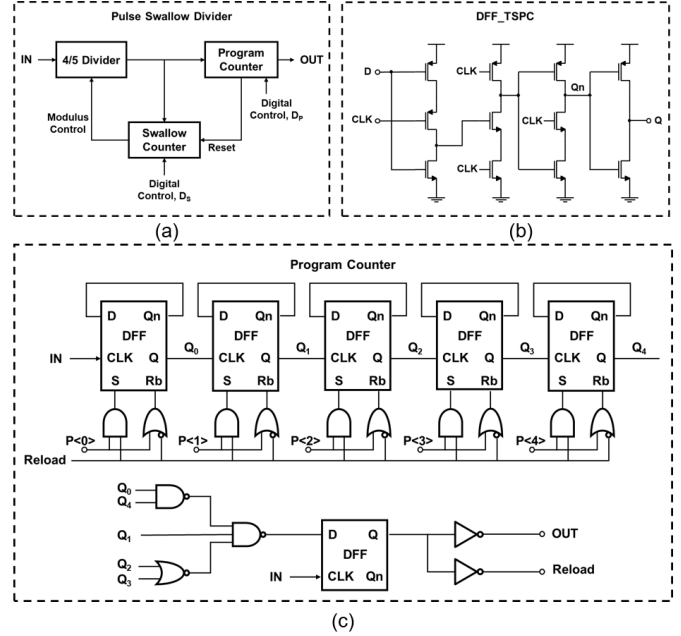


Fig. 5. Circuit implementation of (a) Pulse Swallow Divider, (b) DFF_TSPC, and (c) Program Counter.

phase-difference outputs Q_A and Q_B . These outputs drive the CP and LPF to produce a control voltage V_{out} that adjusts the VCRO and VCRO_TX frequencies identically. In conventional charge-pump PLL (CPPLL) designs, the direct conduction path between CP and LPF causes V_{out} to drift during idle periods. To address this issue, the proposed SCVH module is designed to maintain and periodically refresh the control voltage. During the calibration phase T_1 ($S = 1$), the SCVH remains active, enabling normal PLL operation and voltage updates. During the silent phase T_2 ($S = 0$), the SCVH is switched off, isolating and holding V_{out} . Simultaneously, the PFD and frequency divider are deactivated, the VCRO halts oscillation, and the charge pump ceases all charge/discharge activity, placing the entire PLL in a nearly zero-power silent mode. This configuration eliminates CP charge/discharge currents, reducing overall power consumption to near-zero levels. The controlled voltage V_{out} is simultaneously fed to identical VCRO_TX for the up-conversion of the transmitter. To ensure correct functionality, T_1 must exceed the PLL's locking time to guarantee phase convergence, while T_2 depends on the

drift tolerance of V_{out} and ambient temperature variation. This time-multiplexed PLL design meets the requirements for periodic frequency calibration while significantly reducing average power consumption. It is particularly well-suited for systems such as neural implants, where power budgets are highly constrained and signal transmission occurs intermittently.

B. Circuit Implementation

1) *CP, SCVH*: The Q_A and Q_B signals generated by the PFD are fed into the charge pump to produce the charge/discharge current I_{net} , which in turn regulates the control voltage V_{out} . However, due to the gate-drain overlap capacitance of the switching transistors, the rising and falling pulses from Q_A and Q_B will couple through the gate-drain capacitance, introducing disturbances to V_{out} . To mitigate this clock feedthrough effect, the switching transistors M_1 and M_2 in Fig. 3 are placed at the supply and ground terminals, respectively. Therefore, the total capacitance from nodes X and Y to ground helps suppress the coupling effect before any change occurs at the source nodes of M_3 and M_4 (which act as current sources).

The SCVH module is implemented using a transmission gate composed of PMOS and NMOS transistors, controlled by the signal S . When $S = 1$, the transmission gate is turned on, allowing the current I_{net} to flow into or out of the low-pass filter (LFP). When $S = 0$, the SCVH is turned off, isolating the charge pump from the LFP and enabling voltage holding on V_{out} . To suppress skew caused by the misalignment of the pull-up and pull-down pulses, a transmission gate is inserted at the gate of M_2 and M_6 to introduce a delay equal to that of an inverter, ensuring synchronized pulse arrival.

2) *PFD*: The PFD employs two True Single-Phase Clock (TSPC) D flip-flops (DFFs) and a reset logic to compare the phase and frequency difference between the reference clock F_{ref} and feedback clock F_{back} . DFF_TSPC_R is a modified version of the TSPC DFF that incorporates a reset function to support rapid clearing, as shown in Fig. 4. The TSPC DFF utilizes a single-phase clocking scheme, eliminating the reliance on complementary clock signals typically required by conventional master-slave DFFs. This helps reduce the impact of clock skew on performance. Additionally, the TSPC DFF is well-suited for high-frequency and low-power applications, aligning with our design goals.

3) *Divider*: The frequency divider, with its ratio set to 50, is realized by a Pulse Swallow Divider, as shown in Fig. 5, which is a flexible integer-N frequency divider to achieve programmable division ratios. It combines a dual-modulus prescaler (divide-by-4/5), a swallow counter, and a program counter. The divider begins in the higher division mode (divide-by-5) and remains in this state for 2 cycles defined by the swallow counter. Once the swallow counter resets, the prescaler switches to the lower division mode (divide-by-4) and continues until the program counter completes its count. The total division ratio is determined by $4P + 2$, where P is the program counter value.

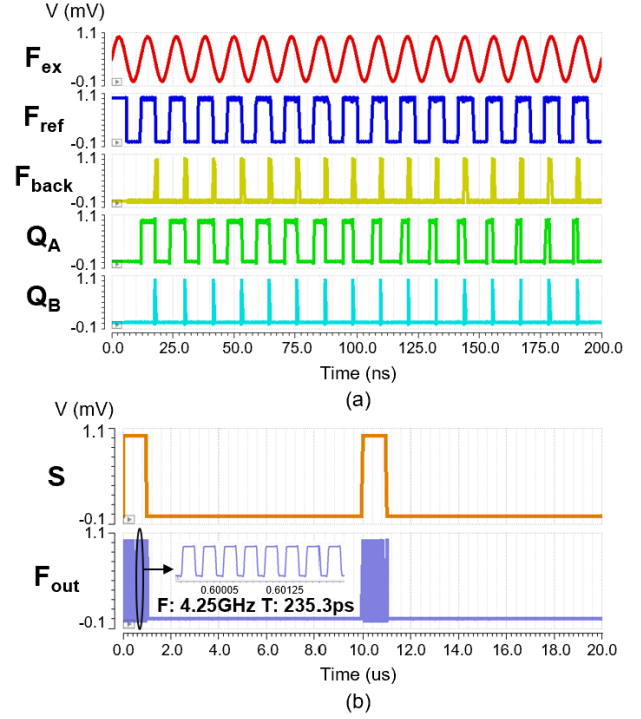


Fig. 6. Waveforms of (a) PFD sensing the input phase difference and (b) PLL's output controlled by signal S .

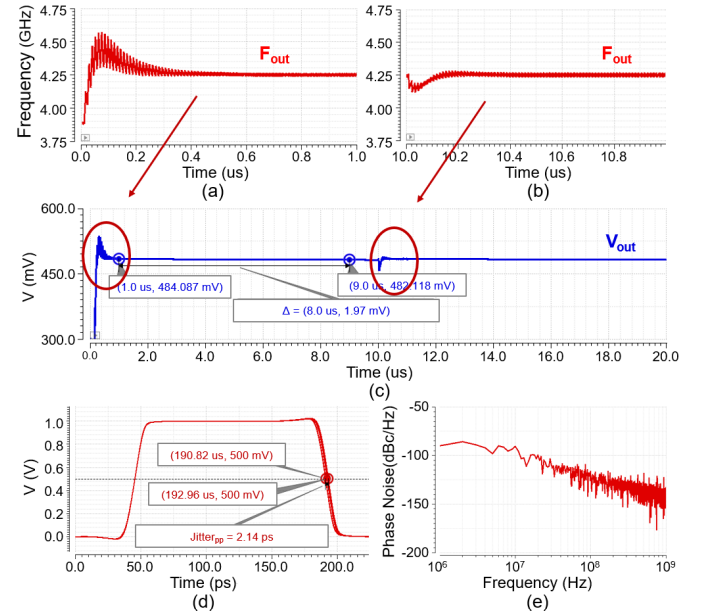


Fig. 7. (a)(b) Frequency curve of PLL output signal. (c) Control voltage V_{out} variation curve of the PLL. (d) Eye diagram of the PLL output signal. (e) Phase noise of PLL.

III. POST-LAYOUT SIMULATION RESULTS

The proposed duty-cycle PLL is designed and implemented in TSMC 40 nm CMOS process technology, with a supply voltage of 1.0 V.

In this study, the carrier frequency of 4.25 GHz, generated by the PLL, is selected due to its lower tissue attenuation

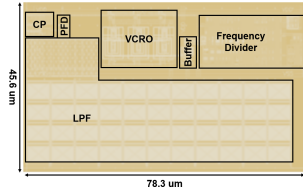


Fig. 8. Layout of the proposed PLL.

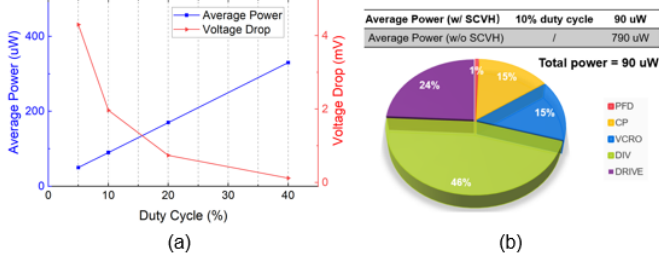


Fig. 9. (a) Average power and voltage drop with different duty cycles. (b) PLL power breakdown.

for the IR-UWB transmitter. The total division ratio of the frequency divider is set to 50. The reference clock signal F_{ref} , generated by shaping external input oscillation F_{ex} , operates at 85 MHz. The control signal S is configured with a period of 10 μ s, where the calibration phase T_1 ($S = 1$) lasts for 1 μ s, corresponding to a 10% duty cycle. Fig. 6 illustrates the frequency difference detected by the PFD between the reference clock and the feedback clock, along with the corresponding PLL output waveform.

As shown in Fig. 7, the system loop enters the locked state within approximately 0.5 μ s of startup. The control voltage V_{out} settles at 484 mV upon locking. During an 8 μ s silent period, the voltage drop due to leakage is less than 2 mV, corresponding to a carrier frequency deviation of less than 0.5%. This deviation is considered negligible within the wide frequency band of the IR-UWB transmitter. As a result, the power consumption of the proposed PLL with the SCVH module is reduced by 89% compared to the design without it. Notably, the voltage-holding mechanism enables faster relocking in subsequent calibration cycles, reducing the relocking time to approximately 0.2 μ s. Additionally, the PLL achieves a locking frequency of 4.25 GHz, confirming correct operation and frequency acquisition functionality. The peak-to-peak jitter ($Jitter_{pp}$) of the PLL output clock is characterized by performing differential sampling of the output signal and analyzing the resulting eye diagram. Using a stable clock signal post-locking, the simulated jitter and phase noise results are shown in Fig. 7(d)(e). With a sampling window of 0.5 μ s and over 2100 clock cycles, the PLL output exhibits a peak-to-peak jitter of 2.14 ps.

As shown in Fig. 8 and Fig. 9, the core area of the proposed PLL occupies only $45.6 \times 78.3 \mu m^2$. Remarkably, it achieves an average power consumption of merely 90 μ W at a duty cycle of 10%. A smaller duty cycle will cause a larger voltage drop, leading to a non-negligible shift in the carrier's center frequency. Table I summarizes the performance of the implemented PLL and provides a comparison with

TABLE I
PERFORMANCE COMPARISON

Parameter	This Work ^b	[18] ^b	[12]	[19]
Process (nm)	40	90	65	65
Supply Voltage (V)	1.0	0.6	1.2	N/A
Frequency (GHz)	4.25	0.915	4.5	2.0
Silent Mode	Yes	Yes	No	No
Jitter _{pp} (ps)	2.14	3.39	N/A	8.03
Core Area (mm ²)	0.0035	0.109	0.16 ^c	0.049
Power (mW)	0.09 ^d	0.2	3.78	9.0
FOM ^a (mW/GHz)	0.021	0.218	0.84	4.5

^aFOM = Power/Frequency (mW/GHz). ^bPost-layout Simulation.

^cEstimated from chip micrograph. ^d10% duty-cycle.

others. While key metrics such as peak-to-peak jitter, core area, and power consumption are included, a normalized metric is needed for a fair assessment. In high-speed circuits like PLLs, dynamic power is a dominant component, scaling nearly linearly with the operating frequency. To account for this inherent dependency, a figure of merit (FOM) is utilized, defined as the ratio of power consumption to operating frequency [18]. This allows for a fairer comparison of architectural efficiency, independent of the target frequency. As shown, the proposed PLL demonstrates a significant reduction in both power consumption and silicon area, achieving a superior FOM compared to its counterparts.

IV. CONCLUSION

In this paper, a low-power, duty-cycled phase-locked loop (PLL) architecture tailored for neural implants is presented. By integrating a switch-controlled voltage holding (SCVH) module, the proposed design enables periodic calibration while maintaining voltage stability during silent phases, achieving very low power consumption while maintaining a stable carrier frequency. The post-layout simulation reveals that the proposed PLL achieves a locking frequency of 4.25 GHz in 0.5 μ s. Operating at a 10% duty cycle, it consumes an average power of 90 μ W, resulting in an 89% power reduction, and exhibits a peak-to-peak jitter of 2.14 ps. Compared to other PLLs, the proposed PLL significantly reduces both power consumption and silicon area, and achieves a superior figure of merit (FOM) compared to its counterparts.

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